

PREAMBLE

```
\documentclass[11pt,a4paper]{article}
[twocolumn] for 23_A1 / 23_C1 style.
fontenc[T1] geometry graphics float
wrapfig
caption subcaption booktabs array
multirow
enumitem xcolor[table] amsmath amssymb
amsfonts amsthm listings algorithm
algpseudocode comment natbib[numbers]
hyperref authblk (hyperref LAST)
\newtheorem{proposition}{Proposition}
```

COMPILE

pdflatex ×2, or with .bib:
 pdflatex → bibtex main → pdflatex ×2
 latexmk -pdf main.tex · -shell-escape for minted

TEXT

```
\textbf \textit \emph bold italic emph
\underline \texttt underline mono
\textsc \textsf smallcaps, sans
\textsl \textrm slanted, roman
\{ \tiny .. \Huge} 10 sizes
“x” ‘x’ quotes
- -- en dash, em dash
\% & \$ \% # \% _ \% { \% } escapes
\\ \\[6pt] break,
break+space
```

```
\noindent \newpage
\clearpage
\hspace{20px} \quad
\hfill
\centering no extra space
```

COLOUR (XCOLOR)

```
\textcolor{red}{x} \colorbox{c}{x}
\definecolor{blue}{RGB}{0,76,153}
teal!70!black = 70% teal + 30% black
```

LISTS (ENUMITEM)

```
itemize / enumerate / description
[label=\arabic*.] 1. [label=(\alph*)]
(a)
[label=(\Alph*)] (A)
[label={(\Roman*)}] (I)
[label=(\roman*)] (i) [label=] no label
[leftmargin=*, itemsep=.5em]
\item[$\dagger$] † \diamond ◇ \oplus ⊕
\rhd ▷ \ast * \star ★ \circ ○ \mid † \ni
⊃
```

MATHS

```
$x$ \[ \] equation equation*
align one number per line
equation+split one number total
align*/equation* none · gather no align pt
& align column · \\ end line
\frac \dfrac \tfrac fractions
\sqrt{x} \sqrt[3]{x} roots
\left( \right) auto size
\lceil \rceil \lfloor \rfloor [ ] [ ]
\left. .. one-sided
\right|_{x=1}
\sum_{i=1}^n
\int_0^\infty
\lim\limits_{x\to a} limits below
\text{if } words in math
\mathbb \mathcal
\mathrm \mathbf
\DeclareMathOperator* argmax
cases: 0, & t<0 \\ e^{-t}, & t\ge 0
Matrices: pmatrix ( ) bmatrix [ ]
Bmatrix { }
\vdots || Vmatrix || || matrix none
\cdots \vdots \ddots
\ref \eqref (adds parens) \pageref
```

THEOREM (AMSTHM)

```
\newtheorem{thm}{Theorem}[section]
\newtheorem{lem}[thm]{Lemma} shared ctr
\begin{proposition}[Name]\label{p:x}
\begin{proof} .. \end{proof} auto □
```

TABLES

```
l c r · p{5cm} wraps · m{ } b{ }
| rule · \hline · \cline{2-4} partial
\toprule \midrule \bottomrule (booktabs)
\multicolumn{2}{c|}{X} merge across
\multirow{3}{*}{X} merge down, then
leave that cell empty + use \cline
\renewcommand{\arraystretch}{1.8}
\rowcolor{gray!25} first in row
\cellcolor{blue!10} first in cell
\rowcolors{2}{gray!10}{white} stripes
Caption ABOVE tables. Label AFTER
caption.
```

FIGURES

```
\includegraphics[width=0.4\linewidth]{f}
scale= height= angle= keepaspectratio
\graphicspath{{figures/}}
[h t b p ! H] — H = float pkg, exactly here
Caption BELOW figures.
subfigure{0.3\textwidth} + \hfill, image
at
width=\linewidth. No blank lines
between!
New row: \par\medskip. Group label:
\textbf{Inner Planets}\par\medskip
wrapfigure{r}{0.4\textwidth} + text after
figure* table* span both columns [t]
```

CODE & ALGORITHMS

```
\lstset{basicstyle=\ttfamily\small,
keywordstyle=\color{customblue}\bfseries,
commentstyle=\color{teal!70!black},
stringstyle=\color{red!70!black},
frame=single, numbers=left,
breaklines=true,
showstringspaces=false,
columns=fullflexible}
\begin{lstlisting}[language=Python,
caption={..}, label={lst:x}]
\lstinputlisting[language=Python]{f.py}
\verb|x| · verbatim · comment
\begin{algorithm}[H]\caption{} \label{}
\begin{algorithmic}[1]
\Require \Ensure \State \Return \gets
\If{} \ElsIf{} \Else \EndIf
\For{} \EndFor \While{} \EndWhile
\Repeat \Until{} \Function{} \EndFunction
\Comment{} · complexity: $T(m)=\Theta(m)$
```

REFS, NOTES, LINKS, CITES

```
\label AFTER \caption; Figure~\ref{}
\footnote{} attaches to preceding char
\footnotemark \footnotetext{} in tables
\href{url}{text} · \url{..}
\hypersetup{colorlinks=true,
urcolor=blue,
citecolor=black, linkcolor=black}
\cite \citep [1] · \citet Author [1]
\citeauthor \citeyear \nocite{*}
\bibliographystyle{unsrnat} (or ieetr,
plain, unsrt, abbrv, plainnat, apalike)
\bibliography{refs} — no .bib extension
or \begin{thebibliography}{9} \bibitem{k}
```

BEAMER

```
\documentclass[aspectratio=169]{beamer}
\usetheme{Madrid}
\usecolortheme{seahorse}
\begin{frame}{Title} / \frametitle{}
[fragile] for code · [shrink=10] [plain]
\pause <3-> <-3> <2-4> <1,3,5> <+>
```

```
\only no space · \uncover \visible keep
space
\onslide<2-> switch · \alt<2>{a}{b}
\setbeamercovered{transparent}
\alert<2>{} \structure<2>{} \textbf<2>{}
block alertblock exampleblock
theorem corollary proof definition lemma
\begin{columns} \column{0.5\textwidth}
\setbeamertemplate{navigation symbols}{}
TIKZ
```

TIKZ

```
\usetikzlibrary{positioning,calc,fit,arrows}
\draw (0,0) – (2,1); every path ends with ;
circle (1) rectangle (2,1) ellipse (2
and 1)
arc[start angle=0,end
angle=30,radius=3mm]
grid – cycle – | – | – . . . controls ..
\fill
\filldraw[fill=green!20,draw=black]
(30:1cm) polar · ++(1,0) move · +(1,0) stay
(a |– b) · ($!0.5!(b)$) midpoint (calc)
-> <- <-> -stealth · -stealth' needs arrows
thin thick very thick ultra thick dashed
\node[block] (a) at (0,0) {txt};
right=of a above=1cm of a (a.north)
\tikzset{block/.style={draw,
fill=blue!10,
rounded corners, minimum width=2.6cm,
minimum height=1cm, align=center}}
\foreach \x in {0,...,4}{ .. }
at (0,{2-\y}) — brace the arithmetic
\begin{scope}[blue] .. \end{scope}
\tikz \draw ..; inline
```

TECHNICAL WRITING

Active voice; always “we”. Short sentences.
 One idea per paragraph; best sentence first.
 Minimise notation. No $\forall \exists \in$ in prose.
 Never start a sentence with a symbol.
 Separate adjacent formulas with words (“where”).
 Define symbols before use. Keep “that”.
 No contractions. Don't overuse commas.
 Capitalise Theorem 1 / Figure 5 / Section 3;
 lowercase “the following lemma”.
 Spell small numbers as adjectives (five vertices);
 numerals as names (increased by 2; 80 embeddings).
 Fact obvious (no proof) · Lemma proves others
 Theorem main result · Corollary immediate.
 Theorem statements must be self-contained.

RESEARCH PAPER

Read: 3 passes. P1 5 C's = Category, Context, Correctness, Contributions, Clarity.
 P2 content, no details. P3 re-create the work.
 Survey: high sensitivity / low specificity.
 Salami science = one study, many papers.
 IMRaD. Intro 5 Qs: what / why interesting / why hard / why unsolved / key components.
 Methods + Results: past tense, no interpretation.
 Prior work: present tense. Figures self-contained.
 Abstract = mini paper; Methods often longest.
 SPJ 7: don't wait write · identify key idea · tell a story · nail contributions · related work later · readers first (examples) · listen to readers.
 Intro = 1 page: problem + contributions only.
 No “rest of this paper is”. Credit is not money.